

NHB supplementary table reproduction proof

Main-text result	Location in this Supplement	Interpretation boundary
Study design (Fig. 1) and Methods	Cohorts and demographics (Section 2), additional Experiment 1 and 4 analyses (Section 3), and the cross-experiment stimulus inventory (Section A; Supplementary Tables 1–3 and 15; Supplementary Figs. 1–3)	Core outcome, FNI and trajectory definitions are in the main Methods; Experiment 1 is a pilot and Experiment 4 is a design-deviation sensitivity
Visual trajectories (Fig. 5a)	Model estimates, planned contrasts and sensitivity analyses (Section 6; Supplementary Tables 9–10)	Primary inference uses the matched Experiment 3/5 cohort; item-aware and all-completer results are sensitivities
Individual profiles (Fig. 2b)	Category-pattern counts and selected-case components (Section 4; Supplementary Figs. 4–5)	Descriptive only; no discrete participant-type claim
Matched FNI associations (Fig. 3)	Reliability, complete association families and sensitivities (Sections 8 and 5.1–5.4; Supplementary Tables 4–5 and 14; Supplementary Figs. 6 and 11a,b)	The FNI is defined in the main Methods; these are Secondary associations, and null results are not proof of independence
Integrated questionnaire associations (Fig. 4)	Complete pooled correlations, the joint questionnaire model, Negative Capability associations and AQ-perceived-benefit associations (Sections 5.5–5.7; Supplementary Figs. 7–9; Supplementary Tables 6–8)	Secondary or exploratory and cross-sectional; pooled coefficients are not measurement-invariant or causal estimates
Trajectory moderators (Fig. 5a–b)	Complete moderator families, item-aware inference and detailed displays (Section 7; Supplementary Tables 11–13; Supplementary Fig. 10)	<i>Exploratory</i> ; Daily-item candidates are model-dependent
Questionnaire battery (Fig. 1c) and Methods	Reconstruction and reliability (Section 8.2; Supplementary Fig. 11c,d), plus provenance, item wording and scoring (Appendix B; Supplementary Table 16)	Measurement implementations differed across experiments; permissions remain a separate publication requirement

Supplementary Table 1. Analysis populations and manuscript roles. Experiment 1 is shown for study chronology as an archived pilot and is excluded from all current analyses; its 15-record archive count is not a frozen cohort denominator. For Experiments 2–5, the QC-clean column denotes the frozen manuscript membership; it does not imply that an identical quality screen was available in every experiment.

Experiment	Setting	Completers	QC-clean	Manuscript role
1	Laboratory	15	–	Pilot; excluded from current analyses
2	Laboratory	24	24	Integrated questionnaire/sample scope
3	Laboratory	22	22	Integrated scope; matched visual analysis
4	Online	126	126	Integrated scope; descriptive target-shift analysis
5	Online	133	131	Integrated scope; matched visual analysis
3 and 5 combined	Mixed	155	153	Matched trial-level cohort
2–5 combined	Mixed	305	303	Primary integrated questionnaire/sample scope

Supplementary Table 2. Experiment-specific visual-task implementations. Experiment 1 is retained as pilot and design-development context, but is excluded from current E2–E5 integrated and inferential analyses. The table separates the shared three-phase logic from implementation differences that affect cross-experiment interpretation.

Study	Setting and blocks	Rating sequence and familiar-target rule	Comparative-preference sequence and manuscript role
1	Laboratory; two sessions across Face, Scenery, Geometry and Car	Nominally 27 ratings; the archived Diagonal-SUV block contains 28; nominal median target (rank 14 of 27)	26 previously encountered comparisons per subcategory; archived pilot, excluded from current analyses
2	Laboratory; eight blocks across Face, Scenery, Geometry and Car	19 rating records per block in all 24 raw exports; target rank not independently code-verified	18 comparisons per block; integrated questionnaire/sample scope only
3	Laboratory; 12 blocks across Face, Scenery and Geometry	19 ratings; nominal median target (rank 10 of 19)	18 comparisons per block; matched visual analysis
4	Online; 12 blocks across Face, Scenery and Geometry	19 ratings; implementation error selected the nominal fifth-highest target (approximately top quartile)	18 comparisons per block; descriptive design-deviation analysis
5	Online; 12 blocks across Face, Scenery and Geometry	19 ratings; corrected median target (rank 10 of 19)	18 comparisons per block; matched visual analysis

Supplementary Table 3. Participant demographics in the primary integrated Experiments 2–5 analysis set.

a. Age and recorded gender at birth

Experiment	Setting	<i>n</i>	Age, mean (SD)	Median [range]	Recorded gender at birth, <i>n</i> (%)
E2	Laboratory	24	25.08 (10.24)	21 [18–62]	Female 14 (58.3%) Male 10 (41.7%)
E3	Laboratory	22	21.91 (6.41)	19 [18–46]	Female 14 (63.6%) Male 8 (36.4%)
E4	Online	126	39.60 (12.65)	36.5 [18–81]	Female 52 (41.3%) Male 74 (58.7%)
E5	Online	131	36.37 (12.16)	35 [19–78]	Female 79 (60.3%) Male 52 (39.7%)
Pooled	Mixed	303	35.77 (13.04)	34 [18–81]	Female 159 (52.5%) Male 144 (47.5%)

b. Recorded race response

Response	E2	E3	E4	E5	Pooled
Asian	19 (79.2%)	12 (54.5%)	14 (11.1%)	20 (15.3%)	65 (21.5%)
Black / African American	0 (0.0%)	0 (0.0%)	13 (10.3%)	21 (16.0%)	34 (11.2%)
Hispanic / Latino	1 (4.2%)	4 (18.2%)	11 (8.7%)	4 (3.1%)	20 (6.6%)
White / Caucasian	4 (16.7%)	6 (27.3%)	76 (60.3%)	78 (59.5%)	164 (54.1%)
Multiracial / Mixed	0 (0.0%)	0 (0.0%)	10 (7.9%)	4 (3.1%)	14 (4.6%)
Other	0 (0.0%)	0 (0.0%)	2 (1.6%)	0 (0.0%)	2 (0.7%)
Prefer not to say	0 (0.0%)	0 (0.0%)	0 (0.0%)	4 (3.1%)	4 (1.3%)

Percentages are within experiment. Age, gender at birth and race response were complete for all 303 participants. Race-response labels follow the task-code categories. No unobserved category is inferred from a zero count. Handedness was available only in the laboratory exports: Right 41/46 (89.1%) and Left 5/46 (10.9%). The online exports did not contain this field. The primary integrated set contains two Experiment 5 exclusions; equivalent quality-control flags were not implemented in every experiment.

Supplementary Table 4. Matched visual-category FNI by questionnaire associations.

Visual FNI	Questionnaire	n	ρ	P	q_{BH}	BH
Face FNI	AQ	153	-.2889	.000293	.00292	yes
Face FNI	Daily familiarity	153	-.0065	.93668	.93668	no
Face FNI	Flow attitude	153	.2020	.01229	.04917	yes
Face FNI	Maemuki	153	.2787	.000487	.00292	yes
Geometry FNI	AQ	153	.0860	.29074	.69778	no
Geometry FNI	Daily familiarity	153	.0734	.36739	.69977	no
Geometry FNI	Flow attitude	153	-.0673	.40820	.69977	no
Geometry FNI	Maemuki	153	-.1508	.06274	.18823	no
Scenery FNI	AQ	153	.0232	.77606	.84661	no
Scenery FNI	Daily familiarity	153	.0568	.48522	.72683	no
Scenery FNI	Flow attitude	153	-.0421	.60569	.72683	no
Scenery FNI	Maemuki	153	-.0436	.59280	.72683	no

Note. Two-sided Spearman correlations in the QC-clean E3/E5 matched cohort. BH adjustment was applied across 12 tests: three visual-category FNIs by AQ, Daily familiarity, Maemuki and historical experiment-specific Flow.

Supplementary Table 5. Matched visual-category FNI by Daily familiarity item associations.

Visual FNI	Daily item	<i>n</i>	ρ	<i>P</i>	<i>q</i> _{BH}	BH
Face FNI	Books/magazines	153	-.0014	.98602	.98602	no
Face FNI	Clothes	153	.0058	.94285	.98602	no
Face FNI	Digital games	153	.0187	.81863	.98602	no
Face FNI	Food/cuisine	153	.0773	.34256	.81561	no
Face FNI	Movies/videos	153	-.1248	.12425	.55304	no
Face FNI	Places/activities	153	-.0041	.95997	.98602	no
Face FNI	Restaurant	153	-.0413	.61237	.95486	no
Face FNI	Snacks/treats	153	.0791	.33092	.81561	no
Geometry FNI	Books/magazines	153	.1423	.07941	.55304	no
Geometry FNI	Clothes	153	-.0366	.65318	.95486	no
Geometry FNI	Digital games	153	-.1204	.13826	.55304	no
Geometry FNI	Food/cuisine	153	.0282	.72968	.97290	no
Geometry FNI	Movies/videos	153	.0340	.67636	.95486	no
Geometry FNI	Places/activities	153	.1515	.06165	.55304	no
Geometry FNI	Restaurant	153	.0092	.90996	.98602	no
Geometry FNI	Snacks/treats	153	.1436	.07649	.55304	no
Scenery FNI	Books/magazines	153	.1336	.09969	.55304	no
Scenery FNI	Clothes	153	-.0182	.82316	.98602	no
Scenery FNI	Digital games	153	-.0674	.40780	.81561	no
Scenery FNI	Food/cuisine	153	.0359	.65945	.95486	no
Scenery FNI	Movies/videos	153	.0353	.66477	.95486	no
Scenery FNI	Places/activities	153	.0723	.37479	.81561	no
Scenery FNI	Restaurant	153	.0728	.37144	.81561	no
Scenery FNI	Snacks/treats	153	.0893	.27206	.81561	no

Note. Two-sided Spearman correlations in the same 153-participant cohort. BH adjustment was applied across 24 tests: three visual-category FNIs by eight Daily familiarity items. No term survived.

Supplementary Table 6. Pooled questionnaire-pair associations.

Measure 1	Measure 2	n	ρ	P	q_{BH}	BH
Maemuki	Flow attitude	303	.6284	1.05×10^{-34}	6.31×10^{-34}	yes
AQ	Maemuki	303	-.5843	3.89×10^{-29}	1.17×10^{-28}	yes
AQ	Flow attitude	303	-.3312	3.46×10^{-9}	6.91×10^{-9}	yes
AQ	Daily familiarity	303	.2345	3.75×10^{-5}	5.62×10^{-5}	yes
Daily familiarity	Maemuki	303	-.2143	.000171	.000206	yes
Daily familiarity	Flow attitude	303	-.0745	.19603	.19603	no

Note. Two-sided Spearman correlations in the primary integrated E2-E5 cohort (n=303). BH adjustment was applied across the six unique pairs among AQ, Daily familiarity, Maemuki and historical experiment-specific Flow.

Supplementary Table 7. Pooled Daily familiarity item by questionnaire associations.

Daily item	Questionnaire	<i>n</i>	ρ	<i>P</i>	<i>q</i> _{BH}	BH
Books/magazines	AQ	303	.1322	.02138	.06415	no
Books/magazines	Flow attitude	303	-.0342	.55289	.66346	no
Books/magazines	Maemuki	303	-.0851	.13953	.22325	no
Clothes	AQ	303	.1014	.07795	.13363	no
Clothes	Flow attitude	303	.0127	.82616	.90127	no
Clothes	Maemuki	303	-.1150	.04541	.10248	no
Digital games	AQ	303	.0702	.22317	.31507	no
Digital games	Flow attitude	303	-.0006	.99104	.99706	no
Digital games	Maemuki	303	-.0274	.63458	.72524	no
Food/cuisine	AQ	303	.2567	6.02×10^{-6}	.000144	yes
Food/cuisine	Flow attitude	303	-.0555	.33537	.44716	no
Food/cuisine	Maemuki	303	-.1669	.00357	.01712	yes
Movies/videos	AQ	303	.1128	.04983	.10248	no
Movies/videos	Flow attitude	303	-.0717	.21341	.31507	no
Movies/videos	Maemuki	303	-.1545	.00706	.02824	yes
Places/activities	AQ	303	.2092	.000246	.00197	yes
Places/activities	Flow attitude	303	-.1101	.05551	.10248	no
Places/activities	Maemuki	303	-.2142	.000172	.00197	yes
Restaurant	AQ	303	.1879	.00101	.00608	yes
Restaurant	Flow attitude	303	-.0002	.99706	.99706	no
Restaurant	Maemuki	303	-.1406	.01431	.04907	yes
Snacks/treats	AQ	303	.1185	.03927	.10248	no
Snacks/treats	Flow attitude	303	-.0404	.48358	.61084	no
Snacks/treats	Maemuki	303	-.1120	.05138	.10248	no

Note. Two-sided Spearman correlations in the primary integrated E2–E5 cohort. BH adjustment was applied across 24 tests: eight Daily familiarity items by AQ, Maemuki and historical experiment-specific Flow. This Flow implementation matches Article Fig. 4a–b and is distinct from the harmonized common-12 Flow score used in the observed-variable model.

Supplementary Table 8. Questionnaire relationships in the joint model.

Analysis sample	Model relationship	<i>n</i>	Standardized coefficient	95% bootstrap CI	95% CI excludes zero?
Primary QC-clean	Daily familiarity → Flow (common 12)	303	.022	[-.080, .121]	no
Primary QC-clean	AQ → Flow (common 12)	303	-.348	[-.448, -.240]	yes
Primary QC-clean	AQ → Maemuki	303	-.398	[-.476, -.322]	yes
Primary QC-clean	Flow (common 12) → Maemuki	303	.514	[.432, .591]	yes
Primary QC-clean	Daily familiarity–AQ correlation	303	.229	[.125, .329]	yes

Note. Standardized coefficients are shown with 95% percentile-bootstrap intervals from 10,000 experiment-stratified participant resamples. Flow is the harmonized common-12 score. Single-headed paths are regression relationships estimated jointly and do not imply causality. The Daily familiarity–AQ row is their standardized correlation and is not directional.

Supplementary Table 9. Primary category trajectories and pairwise differences in the QC-clean matched cohort ($n = 153$; 33,048 trials).

Estimand	Estimate	SE	95% CI	P	q_{BH}
a. Category-specific slopes					
Face	.01337	.00285	[.00778, .01896]	2.73×10^{-6}	—
Scenery	-.01763	.00285	[-.02322, -.01204]	6.26×10^{-10}	—
Geometry	.00131	.00285	[-.00428, .00690]	.646	—
b. Pairwise category-slope differences					
Face – Scenery	.03100	—	[.02310, .03890]	1.49×10^{-14}	4.46×10^{-14}
Face – Geometry	.01206	—	[.00416, .01996]	.00278	.00278
Scenery – Geometry	-.01894	—	[-.02684, -.01104]	2.64×10^{-6}	3.95×10^{-6}

Note. Positive slopes indicate change toward familiarity and negative slopes indicate change toward novelty. Fixed effects use two-sided Wald tests and 95% confidence intervals. BH adjustment applies only to the three planned pairwise differences in panel b.

Supplementary Table 10. Sensitivity analyses for the category trajectories.

Estimand	Estimate	SE	95% CI	<i>P</i>	<i>q</i> _{BH}
a. Item-aware inference in the primary cohort (<i>n</i> = 153)					
Face slope	.01337	.00278	[.00792, .01882]	1.54×10^{-6}	—
Scenery slope	-.01763	.00403	[-.02552, -.00974]	1.19×10^{-5}	—
Geometry slope	.00131	.00391	[-.00636, .00898]	.73785	—
Face – Scenery	.03100	.00476	[.02166, .04034]	7.75×10^{-11}	2.32×10^{-10}
Face – Geometry	.01206	.00451	[.00322, .02090]	.00748	.00748
Scenery – Geometry	-.01894	.00558	[-.02988, -.00799]	.000695	.00104
b. All-completer directional sensitivity (<i>n</i> = 155)					
Face slope	0.01336	0.00726	[-0.00087, 0.02760]	0.0658	—
Scenery slope	-0.01819	0.00305	[-0.02417, -0.01222]	2.44×10^{-9}	—
Geometry slope	0.00112	0.00368	[-0.00610, 0.00834]	0.761	—
Face – Scenery	0.03156	0.00814	[0.01560, 0.04751]	0.000106	0.000159
Face – Geometry	0.01224	0.00870	[-0.00480, 0.02929]	0.159	0.159
Scenery – Geometry	-0.01931	0.00474	[-0.02861, -0.01001]	4.69×10^{-5}	0.000141

Note. Panel a uses CR1 standard errors clustered by participant, familiar stimulus and novel stimulus; its coefficients match the primary model. Panel b uses the richer participant-slope model in the all-completer cohort. BH adjustment applies separately to the three pairwise differences in each panel; slope tests were not included in those families.

Supplementary Table 11. Complete overall-moderator trajectory family and item-aware sensitivity.

Moderator	b	SE_{HLM}	P_{HLM}	q_{HLM}	SE_{CR1}	P_{CR1}	q_{CR1}	CR1	BH
Face									
AQ	-.00046	.00041	.25676	.73238	.00030	.12433	.49731		no
Daily familiarity	.00101	.00285	.72204	.81317	.00271	.70818	.86037		no
Maemuki	.00489	.00476	.30516	.73238	.00463	.29125	.75095		no
Flow Attitude	.00337	.00447	.45059	.77245	.00381	.37548	.75095		no
Scenery									
AQ	.00010	.00041	.80803	.81317	.00056	.86037	.86037		no
Daily familiarity	.00670	.00285	.01879	.11273	.00429	.11807	.49731		no
Maemuki	-.00220	.00476	.64425	.81317	.00701	.75350	.86037		no
Flow Attitude	.00589	.00447	.18761	.73238	.00636	.35427	.75095		no
Geometry									
AQ	-.00010	.00041	.81317	.81317	.00054	.85723	.86037		no
Daily familiarity	-.00245	.00285	.39043	.77245	.00430	.56873	.86037		no
Maemuki	-.00167	.00476	.72627	.81317	.00481	.72871	.86037		no
Flow Attitude	-.01426	.00447	.00143	.01712	.00467	.00224	.02684		yes

Note. Coefficients are moderator by category by comparison-position interactions. HLM q values use BH across the 12 tests displayed in main-text Figure 5b; CR1 uses the item-aware covariance described in Section 5.2 and BH across the same 12 tests. AQ, Maemuki and historical experiment-specific Flow retain their original raw-score units; the eight-item Daily familiarity composite was standardized before fitting.

Supplementary Table 12. Joint-trait trajectory-moderation sensitivity.

Trait	<i>b</i>	SE	<i>z</i>	<i>P</i>	<i>q</i> _{BH}	BH
Face						
AQ	-.00032	.00053	-.614	.53944	.91819	no
Maemuki	.00099	.00765	.129	.89709	.91819	no
Flow Attitude	.00061	.00590	.103	.91819	.91819	no
Negative Capability	.00138	.00464	.297	.76643	.91819	no
Scenery						
AQ	-.00007	.00053	-.140	.88883	.91819	no
Maemuki	-.01103	.00765	-1.442	.14927	.35824	no
Flow Attitude	.01103	.00590	1.870	.06143	.18429	no
Negative Capability	.00131	.00464	.283	.77750	.91819	no
Geometry						
AQ	-.00029	.00053	-.549	.58313	.91819	no
Maemuki	.01540	.00765	2.014	.04404	.18429	no
Flow Attitude	-.01802	.00590	-3.054	.00226	.02708	yes
Negative Capability	-.00886	.00464	-1.910	.05609	.18429	no

Note. AQ, Maemuki, historical experiment-specific Flow and Negative Capability were entered together. BH adjustment was applied across the 12 joint-model category-slope interactions. The joint model is a secondary collinearity sensitivity and was not used to select the overall-moderator result.

Supplementary Table 13. Complete exploratory Daily-item moderation family with original mixed-model and item-aware sensitivity inference.

Daily item	<i>b</i>	SE _{HLM}	<i>P</i> _{HLM}	<i>q</i> _{HLM}	SE _{CR1}	<i>P</i> _{CR1}	<i>q</i> _{CR1}	CR1 BH
Face								
Food/cuisine	.00039	.00285	.89090	.96814	.00262	.88141	.97729	no
Places/activities	.00026	.00285	.92611	.96814	.00244	.91382	.97729	no
Movies/videos	-.00321	.00285	.26006	.62415	.00258	.21370	.59905	no
Restaurant	.00110	.00285	.69839	.89310	.00251	.65952	.97729	no
Clothes	.00352	.00285	.21633	.57689	.00236	.13535	.59905	no
Digital games	.00018	.00285	.94959	.96814	.00321	.95520	.97729	no
Books/magazines	.00058	.00285	.83787	.96814	.00296	.84395	.97729	no
Snacks/treats	.00259	.00285	.36285	.72570	.00262	.32197	.70248	no
Scenery								
Food/cuisine	.00151	.00285	.59548	.89310	.00453	.73807	.97729	no
Places/activities	.00513	.00285	.07177	.33773	.00372	.16796	.59905	no
Movies/videos	-.00116	.00285	.68399	.89310	.00408	.77594	.97729	no
Restaurant	.00969	.00285	.000677	.00812	.00378	.01049	.12586	no
Clothes	.00011	.00285	.96814	.96814	.00400	.97729	.97729	no
Digital games	.00492	.00285	.08443	.33773	.00424	.24601	.59905	no
Books/magazines	.00243	.00285	.39356	.72656	.00440	.58045	.97729	no
Snacks/treats	.01156	.00285	4.96×10^{-5}	.00119	.00378	.00220	.05279	no
Geometry								
Food/cuisine	.00418	.00285	.14248	.42745	.00359	.24359	.59905	no
Places/activities	.00107	.00285	.70703	.89310	.00388	.78272	.97729	no
Movies/videos	-.00431	.00285	.13081	.42745	.00374	.24961	.59905	no
Restaurant	-.00571	.00285	.04501	.27004	.00364	.11687	.59905	no
Clothes	-.00628	.00285	.02753	.22023	.00367	.08697	.59905	no
Digital games	-.00287	.00285	.31340	.68378	.00323	.37369	.74738	no
Books/magazines	.00116	.00285	.68506	.89310	.00306	.70601	.97729	no
Snacks/treats	.00212	.00285	.45737	.78406	.00362	.55795	.97729	no

Note. Each coefficient is the standardized Daily-item by category by comparison-position interaction in the matched E3/E5 cohort (153 participants; 33,048 trials). Original HLM *q* values use BH across 24 tests. CR1 uses participant-by-category fixed effects, subcategory fixed effects, and multiway clustering over participant, familiar stimulus and novel stimulus, with BH across the same 24 coefficients. The fixed coefficient is unchanged; only the covariance estimator differs. HLM denotes the selected converged mixed-model fallback. CR1 BH indicates whether $q < .05$.

Supplementary Table 14. Cross-subcategory stability. Entries are median Spearman–Brown coefficients with 2.5th–97.5th percentiles across 10,000 balanced subcategory partitions.

Measure	Face	Scenery	Geometry
Category-specific FNI	0.235 [0.061,0.396]	0.469 [0.348,0.579]	0.323 [0.157,0.473]
Category-specific trajectory slope	-0.322 [-0.601,-0.060]	0.068 [-0.157,0.270]	0.071 [-0.150,0.268]

Supplementary Table 15. Complete task-pool stimulus taxonomy across Experiments 1–5. A solid dot indicates availability and a dash indicates absence. All 16 Face families were available in every experiment; participant-level assignment is described below.

Category and harmonized subcategory	E1	E2	E3	E4	E5
Face					
Female–Asian–Adult	•	•	•	•	•
Female–Asian–Young-adult	•	•	•	•	•
Female–Black–Adult	•	•	•	•	•
Female–Black–Young-adult	•	•	•	•	•
Female–Latino–Adult	•	•	•	•	•
Female–Latino–Young-adult	•	•	•	•	•
Female–White–Adult	•	•	•	•	•
Female–White–Young-adult	•	•	•	•	•
Male–Asian–Adult	•	•	•	•	•
Male–Asian–Young-adult	•	•	•	•	•
Male–Black–Adult	•	•	•	•	•
Male–Black–Young-adult	•	•	•	•	•
Male–Latino–Adult	•	•	•	•	•
Male–Latino–Young-adult	•	•	•	•	•
Male–White–Adult	•	•	•	•	•
Male–White–Young-adult	•	•	•	•	•
Scenery					
Beach	•	–	•	•	•
Desert	•	–	•	•	•
Flower	•	–	–	–	–
Lake	•	–	–	–	–
Mountain	•	•	•	•	•
River	•	•	•	•	•
Sky	•	–	–	–	–
Geometry					
Complex (symmetry not encoded)	•	•	–	–	–
Simple (symmetry not encoded)	•	•	–	–	–
Complex–asymmetric	–	–	•	•	•
Complex–symmetric	–	–	•	•	•
Simple–asymmetric	–	–	•	•	•
Simple–symmetric	–	–	•	•	•
Car					
Diagonal sedan	•	–	–	–	–
Diagonal sports car	•	•	–	–	–
Diagonal SUV	•	•	–	–	–
Diagonal truck	•	–	–	–	–
Front sedan	•	–	–	–	–
Front sports car	•	–	–	–	–
Front SUV	•	–	–	–	–
Front truck	•	–	–	–	–

Supplementary Table 16. Detailed questionnaire-derived features used in Experiments 2–5. Negative Capability was administered only in Experiments 3–5.

Feature	Construct	Items / Source	Scoring	Interpretation
1_Trust-Integrity	Trust and integrity orientation	11 trust-related items	Mean of 11 Likert items; standard items scored –3 to +3; reverse-coded items reversed before averaging	Higher = stronger trust in others, honesty, kindness, and interpersonal integrity
2_Emotion-Coping	Emotion regulation and coping	12 emotion/coping items	Mean of 12 Likert items after reverse coding	Higher = stronger adaptive emotion regulation and coping tendency
3_Social-self	Social self-expression and interpersonal self-orientation	15 social/self-expression items	Mean of 15 Likert items after reverse coding	Higher = stronger social openness, self-expression, and interpersonal engagement
4_food/cuisine	Daily novelty/familiarity in food choices	1 daily-life behavior item	–3 = Always New, 0 = Intermediate, +3 = Familiar	Higher = more familiar food choices; lower = more novel food choices
4_places/activities	Daily novelty/familiarity in places and activities	1 daily-life behavior item	–3 to +3 novelty/familiarity coding	Higher = more familiar places/activities; lower = more novel places/activities
4_movies or videos	Daily novelty/familiarity in media choices	1 daily-life behavior item	–3 to +3 novelty/familiarity coding	Higher = more familiar movies/videos; lower = more novel media choices
4_restaurant	Daily novelty/familiarity in restaurant choices	1 daily-life behavior item	–3 to +3 novelty/familiarity coding	Higher = more familiar restaurant choices; lower = more novel restaurant choices
4_clothes	Daily novelty/familiarity in clothing choices	1 daily-life behavior item	–3 to +3 novelty/familiarity coding	Higher = more familiar clothing choices; lower = more novel clothing choices
4_digital games	Daily novelty/familiarity in digital games	1 daily-life behavior item	–3 to +3 novelty/familiarity coding	Higher = more familiar game choices; lower = more novel game choices
4_books or magazines	Daily novelty/familiarity in reading choices	1 daily-life behavior item	–3 to +3 novelty/familiarity coding	Higher = more familiar reading choices; lower = more novel reading choices
4_snacks or treats	Daily novelty/familiarity in snack choices	1 daily-life behavior item	–3 to +3 novelty/familiarity coding	Higher = more familiar snack choices; lower = more novel snack choices
4_NF-daily	Overall daily novelty/familiarity tendency	Mean of 8 daily-life domains	Mean of the 8 4_ daily behavior items	Higher = stronger familiarity-seeking in daily life; lower = stronger novelty-seeking
4_1_N-affect_daily	Self-reported lifestyle impact of novelty-seeking	2 past-six-month items: novelty-seeking made life more positive/active; novelty-seeking made life more resilient against negative situations	Mean of the 2 novelty-seeking impact items scored –3 to +3	Higher = stronger self-reported positive lifestyle impact of novelty-seeking
4_2_F-affect_daily	Self-reported lifestyle impact of familiarity-seeking	2 past-six-month items: familiarity-seeking made life more positive/active; familiarity-seeking made life more resilient against negative situations	Mean of the 2 familiarity-seeking impact items scored –3 to +3	Higher = stronger self-reported positive lifestyle impact of familiarity-seeking

Supplementary Table 16. Detailed questionnaire-derived features used in Experiments 2–5 (continued).

Feature	Construct	Items / Source	Scoring	Interpretation
5_Social-oppo-group	Social opportunity and group mobility	12 items about meeting people, social options, and group choice	Mean of 12 Likert items after reverse coding	Higher = greater perceived social opportunity and freedom to choose social groups
6_Other-social	Perceived social environment of others	12 items about people the respondent regularly interacts with	Mean of 12 Likert items after reverse coding	Higher = respondent perceives surrounding others as having more social opportunity and mobility
7_Personal_resp-self_disp	Personal responsibility and self-discipline	19 responsibility/self-discipline items	Mean of 19 Likert items after reverse coding	Higher = stronger responsibility, persistence, and self-discipline
8_Communi_Family	Community and family engagement	14 community/family items	Mean of 14 Likert items scored –3 to +3	Higher = stronger family/community connection and engagement
9_Growth_Reflect	Personal growth and self-reflection	16 personal growth/reflection items	Mean of 16 Likert items after reverse coding	Higher = stronger growth orientation and reflective tendency
10_Flow-attitude	General flow orientation	12 flow-related items	Mean of 12 Likert items after reverse coding	Higher = stronger positive attitude, desire, and tendency toward flow-like experiences
10_Flow_belief_tendency	Belief/value placed on flow	First 4 flow items	Mean of 4 scored flow items	Higher = stronger belief that flow is valuable or meaningful
10_Flow_desire_tendency	Desire for flow	One flow-related desire item	Single Likert item scored –3 to +3	Higher = stronger desire to experience flow
10_Flow_real_tendency	Actual flow experience tendency	2 flow-experience items	Mean of 2 scored flow items	Higher = more reported real-life flow experience
11_Self_percept-worth	Self-perception and self-worth	10 self-worth items	Mean of 10 Likert items after reverse coding	Higher = stronger positive self-perception and self-worth
12_Negative_Capability	Negative capability / adaptive tolerance of uncertainty	9 negative-capability items; Experiments 3–5 only	Mean of 9 Likert items scored –3 to +3	Higher = stronger ability to tolerate uncertainty, difficulty, or ambiguity and respond adaptively
AQ_score	Autism-spectrum trait score	50 Autism-Spectrum Quotient items	Binary scoring per item, then summed; each item scored 0 or 1	Higher = stronger autistic-trait tendency
Maemuki_score	Broad positive/forward psychological orientation	Composite of 9 subscales: 1, 2, 3, 5, 6, 7, 8, 9, and 11	Mean of the 9 questionnaire subscale scores	Higher = more positive, socially engaged, self-directed, and forward-looking psychological profile

Measure	Adjusted r	q_{BH}	Joint β	P
AQ	-0.250	0.0132	-0.143	0.158
Maemuki	0.263	0.0132	0.115	0.384
Flow	0.208	0.0388	0.087	0.412